GOV 51 PS3

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GitHub Repository: https://github.com/jennyjia319/gov51-ps3

1 Setup and Data Exploration

1.1 Load Packages and Data

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(glmnet)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 4.1-10
```

```r
library(knitr)
compas <- read.csv("data/compas_clean.csv")
```

1.2 How many observations and variables are in the dataset? What is the unit of observation?

```r
dim(compas)
```

```
## [1] 7214   17
```

```r
head(compas)
```

```
##   recidivism age female black hispanic asian other_race native_american
## 1          0  69      0     0        0     0          1               0
## 2          1  34      0     1        0     0          0               0
## 3          1  24      0     1        0     0          0               0
## 4          0  23      0     1        0     0          0               0
## 5          0  43      0     0        0     0          1               0
## 6          0  44      0     0        0     0          1               0
##   priors_count felony juv_fel_count juv_misd_count juv_other_count
## 1            0      1             0              0               0
## 2            0      1             0              0               0
## 3            4      1             0              0               1
## 4            1      1             0              1               0
## 5            2      1             0              0               0
## 6            0      0             0              0               0
##   has_juv_felony has_juv_misd age_under_25 age_25_45
## 1              0            0            0         0
## 2              0            0            0         1
## 3              0            0            1         0
## 4              0            1            1         0
## 5              0            0            0         1
## 6              0            0            0         1
```

There are 7214 observations and 17 variables in this dataset. The unit of observation is defendants, since each row includes demographics and the criminal history of an individual.

1.3 Examine the Outcome Variable

```r
mean(compas$recidivism)
```

```
## [1] 0.4506515
```

The overall recidivism rate is 45.06515%, which means about 45% of the individuals were rearrested. The outcome is binary because it takes the value of either 0 or 1. This is an important consideration because predictions made from a linear model are interpreted as probabilities, so the model can produce values that are outside 0-1, which are invalid probabilities that are less fit for binary variables.

Summary Statistics

```r
compas |>
  summarise(across(everything(),
                   list(mean = mean, sd = sd),
                   na.rm = TRUE))
```

```
## Warning: There was 1 warning in `summarise()`.
## i In argument: `across(everything(), list(mean = mean, sd = sd), na.rm =
##   TRUE)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
##   recidivism_mean recidivism_sd age_mean   age_sd female_mean female_sd
## 1       0.4506515     0.4975933 34.81799 11.88892    0.193374  0.394971
##   black_mean  black_sd hispanic_mean hispanic_sd  asian_mean   asian_sd
## 1  0.5123371 0.4998824    0.08830053   0.2837511 0.004435819 0.06645867
##   other_race_mean other_race_sd native_american_mean native_american_sd
## 1       0.0522595     0.2225653          0.002495148         0.04989256
##   priors_count_mean priors_count_sd felony_mean felony_sd juv_fel_count_mean
## 1          3.472415        4.882538   0.6467979  0.477998         0.06723039
##   juv_fel_count_sd juv_misd_count_mean juv_misd_count_sd juv_other_count_mean
## 1        0.4739715          0.09093429         0.4852385            0.1093707
##   juv_other_count_sd has_juv_felony_mean has_juv_felony_sd has_juv_misd_mean
## 1          0.5015858          0.03909066         0.1938241        0.05752703
##   has_juv_misd_sd age_under_25_mean age_under_25_sd age_25_45_mean age_25_45_sd
## 1        0.232863          0.211949       0.4087172      0.5695869    0.4951683
```

Variables like priors_count and age have the most variation, as shown by their high standard deviations. On the other hand, variables like native_american and asian show the least variation, with standard deviations close to zero. What this tells us is that their values are mostly concentrated around the binary value of 0 for the observations.

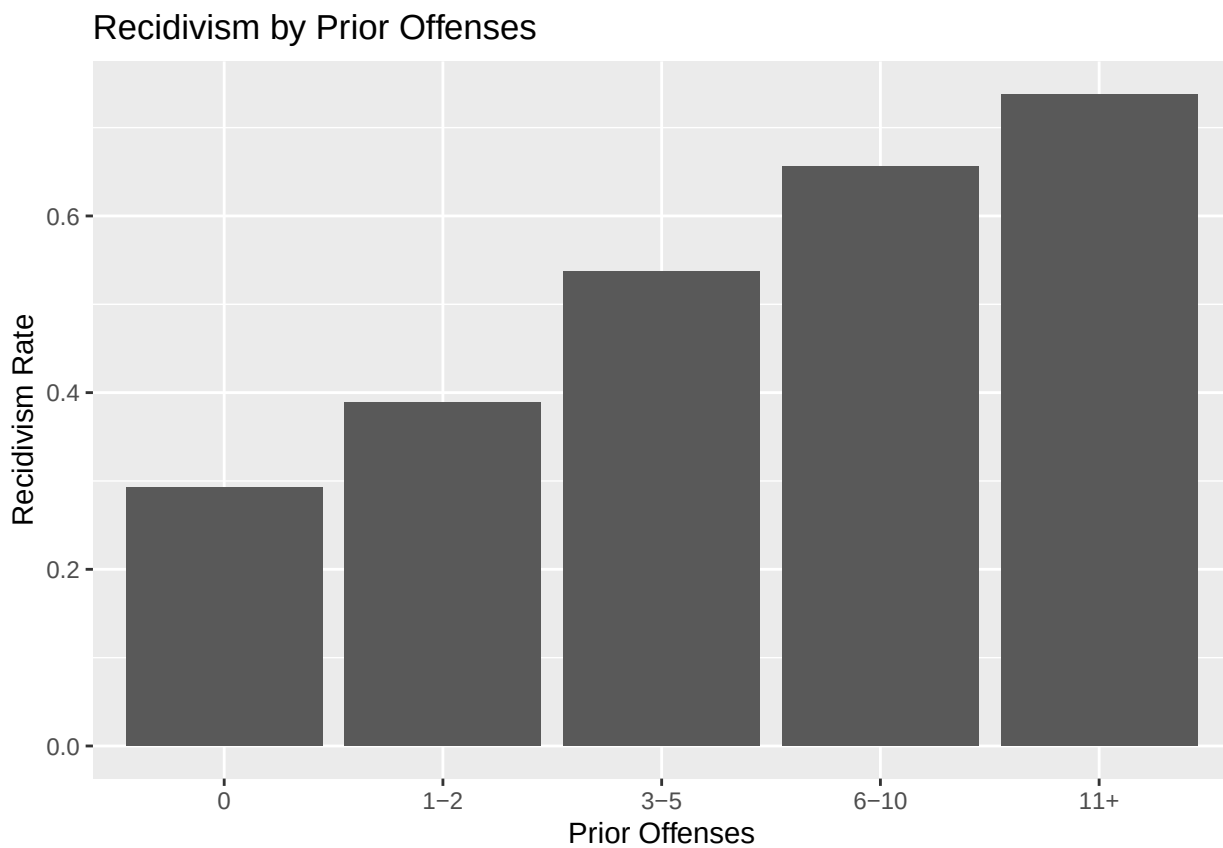
1.5 Explore Key Predictors

```r
compas |>
  mutate(priors_groupped = case_when(
    priors_count == 0 ~ "0",        # no prior offenses
    priors_count <= 2 ~ "1-2",
    priors_count <= 5 ~ "3-5",
    priors_count <= 10 ~ "6-10",
    TRUE ~ "11+"
  )) |>
mutate(priors_groupped = factor(priors_groupped,
                          levels = c("0", "1-2", "3-5", "6-10", "11+"))) |> #  prior offenses years wer
```

```r
group_by(priors_groupped) |> # Group by the new categories |>

summarise(rate = mean(recidivism)) |> # Calculate mean recidivism rate |>


ggplot(aes(x = priors_groupped, y = rate)) +
geom_col() +
labs(x = "Prior Offenses",
     y = "Recidivism Rate",
     title = "Recidivism by Prior Offenses")
```



What we observe from the graph is that there is indeed a positive relationship between prior offenses and recidivism rates. This information suggests that prior offenses could be a good predictor of repeat offenses/recidivism.

OLS Baseline

2.1 Train/Test Split

```r
set.seed(51)
n <- nrow(compas)
train_idx <- sample(1:n, size = floor(0.8 * n))
train <- compas[train_idx, ]
```

```r
test <- compas[-train_idx, ]

nrow(train)
```

```
## [1] 5771
```

```r
nrow(test)
```

```
## [1] 1443
```

There are 5771 observations in the training set, and 1443 observations in the test set.

2.2 Fit OLS with Main Effects Only

```r
ols <- lm(recidivism ~ ., data = train)
summary(ols)
```

```
##
## Call:
## lm(formula = recidivism ~ ., data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3347 -0.3966 -0.1892  0.4520  1.1865
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.732659   0.065429  11.198  < 2e-16 ***
## age             -0.010259   0.001169  -8.772  < 2e-16 ***
## female          -0.059808   0.015632  -3.826 0.000132 ***
## black            0.012266   0.014002   0.876 0.381049
## hispanic        -0.031099   0.023196  -1.341 0.180073
## asian           -0.035871   0.092911  -0.386 0.699452
## other_race      -0.046504   0.028482  -1.633 0.102577
## native_american -0.020283   0.119810  -0.169 0.865572
## priors_count     0.029532   0.001401  21.072  < 2e-16 ***
## felony           0.037706   0.012938   2.914 0.003578 **
## juv_fel_count    0.005821   0.017516   0.332 0.739647
## juv_misd_count  -0.028568   0.018950  -1.508 0.131723
## juv_other_count  0.023089   0.012614   1.830 0.067244 .
## has_juv_felony   0.041631   0.044485   0.936 0.349395
## has_juv_misd     0.081995   0.039875   2.056 0.039802 *
## age_under_25    -0.015671   0.040972  -0.382 0.702119
## age_25_45       -0.081454   0.028916  -2.817 0.004865 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4613 on 5754 degrees of freedom
## Multiple R-squared:  0.1429, Adjusted R-squared:  0.1405
## F-statistic: 59.95 on 16 and 5754 DF,  p-value: < 2.2e-16
```

(a) At the 5% level, the statistically significant predictors are age, female, black, priors_count, felony, has_juvy_misd, and age_25_45.

(b) To interpret priors_count coefficient: A one unit increase in priors_count is associate with a 0.029532 increase in the predicted probability of recidivism.

2.3 In-sample vs Out-of-sample RMSE

```
train_data <- predict(ols, newdata = train) #  (a) training data predictions

rmse_train <- sqrt(mean((train_data - train$recidivism)^2))
rmse_train
```

```
## [1] 0.4605811
```

```
test_data <- predict(ols, newdata = test) # (b) test data predictions

rmse_test <- sqrt(mean((test_data - test$recidivism)^2))
rmse_test
```

```
## [1] 0.462227
```

```
rmse_test - rmse_train # (c) gap calculation
```

```
## [1] 0.001645971
```

(a) (b) The in-sample (train) RMSE is 0.4606, while the out-of-sample (test) is 0.4622, and so the gap is 0.001645971.

(b) This gap is very small, which means the model performs similarly on both the training and test data. Thus, there is minimal evidence to suggest that OLS could be overfitting with the 16 predictors.

2.4 Cross-validation By Hand

```
set.seed(51)
folds <- sample(rep(1:5, length.out = nrow(train))) # first need to run this
rmse_cv <- numeric(5)

for (k in 1:5) {
  train_k <- train[folds != k, ]
  test_k  <- train[folds == k, ]

  model_k <- lm(recidivism ~ ., data = train_k)
  pred_k  <- predict(model_k, newdata = test_k)

  rmse_cv[k] <- sqrt(mean((test_k$recidivism - pred_k)^2))
}

mean(rmse_cv)
```

```
## [1] 0.4620144
```

(a) The 5-fold cross-validation RMSE is 0.4620144, which is essentially identical to the single-split out-of-sample RMSE of 0.462227. This indicates that the model's performance is stable and is not sensitive to whether or how the data is split.

(b) I think cross-validation is more reliable because it assesses performance over multiple data splits rather than just one. This reduces random variation, and gives a better estimate of out-of-sample performance.

(c) Cross-validation is best for choosing lambda as it tests how values perform on unseen data and minimizes prediction error.

3 The Kitchen Sink: What Happens When You Add Too Many Predictors?

#.1 Build the Expanded Model

```r
# Full interaction formula   # run according to pset
f_full <- recidivism ~ (.)^2 +
I(age^2) + I(priors_count^2) +
I(juv_fel_count^2) + I(juv_misd_count^2) +
I(juv_other_count^2)

# Create design matrices for glmnet (we'll need these later)
X_train <- model.matrix(f_full, data = train)[, -1]
X_test <- model.matrix(f_full, data = test)[, -1]
y_train <- train$recidivism
y_test <- test$recidivism

# How many features now?
ncol(X_train)
```

```
## [1] 141
```

(a) The expanded model has 141 predictors.
(b) The ratio of training observations to predictors is 40.929078:1 (since 5771/141 = 40.929078). On the other hand, the simple model ratio is 360.6875:1 (since 5771.16 = 360.6875). This is a drastic increase in predictors and model complexity, which raises the risk of overfitting.

3.2 Fit OLS with the Kitchen Sink

```r
ols_full <- lm(f_full, data = train)

train_data_full <- predict(ols_full, newdata = train) # (a) in-sample RMSE

rmse_train_full <- sqrt(mean((train_data_full - train$recidivism)^2))
rmse_train_full
```

```
## [1] 0.4485474
```

```r
test_data_full <- predict(ols_full, newdata = test) # (b) out-of-sample RMSE
```

```
## Warning in predict.lm(ols_full, newdata = test): prediction from rank-deficient
## fit; attr(*, "non-estim") has doubtful cases
```

```r
rmse_test_full <- sqrt(mean((test_data_full - test$recidivism)^2))
rmse_test_full
```

```
## [1] 0.4890411
```

```r
rmse_test_full - rmse_train_full # (c) gap comparison
```

```
## [1] 0.04049367
```

(a) The in-sample RMSE is 0.4485474.
(b) The out-of-sample RMSE is 0.4890411.
(c) The gap between the in-sample and out-of-sample RMSE is 0.04049367, which is much larger than in the simple model of 0.001645971. This indicates significant overfitting in that, while the expanded model achieves a lower in-sample RMSE, its out-of-sample performance The inclusion of a large number of predictors, including interaction and higher-order terms, greatly increases model complexity, causing the model to fit noise in the training data rather than underlying patterns, thereby reducing its ability to generalize to new data.

3.3 The Overfitting Lesson

Adding many interaction terms increases the model's flexibility, which allows it to fit the training data extremely closely and bring the R square value closer to near-1. However, the problem is that this improved in-sample fit is not particularly meaningful, since the model begins to capture noise rather than true patterns, leading to worse performance on new data. This phenomenon is called overfitting.

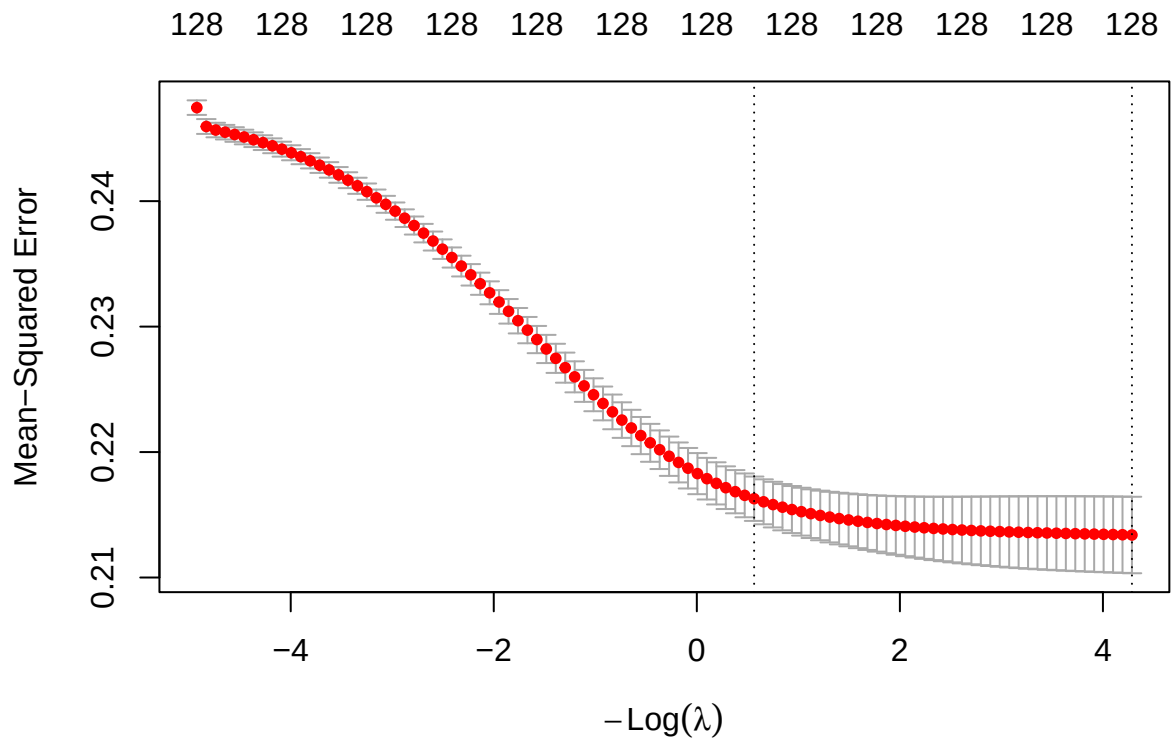
4 Ridge Regression: Shrink Everything

4.1 Fit Ridge with Cross-Validation

```r
set.seed(51)
cv_ridge <- cv.glmnet(X_train, y_train, alpha = 0, nfolds = 10)
cv_ridge$lambda.min
```

```
## [1] 0.01376272
```

```r
plot(cv_ridge)
```



(a) The optimal lambda is 0.01376272. (b) The plot shows how prediction error changes for different values of lambda. We can observe that, as lambda increases, the model becomes more complex with less shrinkage, while smaller lambda values make the model simpler. The lowest point on the curve shows the lambda that gives the best prediction performance with the lowest mean squared error.

4.2 Ridge RMSE

```
ridge_predicted <- predict(cv_ridge, newx = X_test, s = "lambda.min")

rmse_ridge <- sqrt(mean((ridge_predicted - y_test)^2))
rmse_ridge
```

```
## [1] 0.4700385
```

The out-of-sample RMSE for Ridge regression is 0.4700385, which is lower than the kitchen sink OLS of 0.4890411 but higher than the simple OLS of 0.462227. This demonstrates that Ridge helps reduce overfitting in more complex models, but it does not outperform the simpler OLS model.

4.3 Did Ridge Set Any Coefficients to Zero?

```
coef(cv_ridge, s = "lambda.min")
```

```
## 142 x 1 sparse Matrix of class "dgCMatrix"
##                                    lambda.min
## (Intercept)                      6.240172e-01
## age                             -7.146883e-03
## female                          -6.310304e-02
## black                            8.833220e-02
## hispanic                        -9.399273e-02
## asian                           -9.631129e-02
## other_race                       1.172025e-02
## native_american                 -3.951980e-01
## priors_count                     2.814853e-02
## felony                           3.058693e-02
## juv_fel_count                    1.076914e-02
## juv_misd_count                   4.656344e-03
## juv_other_count                 -1.444343e-03
## has_juv_felony                   1.081862e-01
## has_juv_misd                     9.650399e-02
## age_under_25                     8.001654e-02
## age_25_45                       -1.704785e-02
## I(age^2)                        -8.791274e-06
## I(priors_count^2)               -7.997212e-04
## I(juv_fel_count^2)               1.229990e-03
## I(juv_misd_count^2)             -5.644604e-04
## I(juv_other_count^2)            -9.162454e-03
## age:female                      -5.157715e-04
## age:black                       -1.586003e-03
## age:hispanic                     1.410240e-03
## age:asian                       -6.709189e-04
## age:other_race                  -9.495021e-04
## age:native_american             -4.925668e-03
## age:priors_count                 2.275284e-04
## age:felony                      -9.447363e-04
## age:juv_fel_count               -6.818760e-04
## age:juv_misd_count              -8.816889e-04
## age:juv_other_count              5.954111e-04
## age:has_juv_felony              -3.010842e-03
## age:has_juv_misd                 1.839858e-03
## age:age_under_25                -5.473185e-03
## age:age_25_45                   -2.113682e-03
## female:black                    -5.557765e-02
## female:hispanic                  1.584091e-02
## female:asian                    -5.213759e-02
## female:other_race                1.963641e-02
## female:native_american           5.948221e-01
## female:priors_count              5.893477e-03
## female:felony                    5.244157e-02
## female:juv_fel_count             1.769619e-01
## female:juv_misd_count            4.550161e-02
## female:juv_other_count          -3.235086e-02
## female:has_juv_felony           -2.090400e-01
## female:has_juv_misd             -1.424213e-01
## female:age_under_25             -7.660951e-02
## female:age_25_45                 4.893753e-02
## black:hispanic                   .
```

```
## black:asian                      .
## black:other_race                 .
## black:native_american            .
## black:priors_count                2.533953e-04
## black:felony                     -2.439377e-03
## black:juv_fel_count               5.469707e-03
## black:juv_misd_count             -1.509481e-02
## black:juv_other_count             7.799527e-02
## black:has_juv_felony              2.720883e-02
## black:has_juv_misd               -1.319078e-01
## black:age_under_25                5.796655e-02
## black:age_25_45                  -4.565325e-02
## hispanic:asian                   .
## hispanic:other_race              .
## hispanic:native_american         .
## hispanic:priors_count            -4.246163e-03
## hispanic:felony                  -4.338215e-03
## hispanic:juv_fel_count            7.029226e-02
## hispanic:juv_misd_count           1.920545e-01
## hispanic:juv_other_count         -2.622615e-02
## hispanic:has_juv_felony           1.554805e-01
## hispanic:has_juv_misd            -5.198679e-01
## hispanic:age_under_25             1.107656e-01
## hispanic:age_25_45                1.912834e-02
## asian:other_race                 .
## asian:native_american            .
## asian:priors_count                1.070921e-02
## asian:felony                      3.956993e-01
## asian:juv_fel_count              .
## asian:juv_misd_count              1.513275e-01
## asian:juv_other_count             5.299783e-01
## asian:has_juv_felony             .
## asian:has_juv_misd                1.545502e-01
## asian:age_under_25               -3.048093e-01
## asian:age_25_45                  -3.728278e-01
## other_race:native_american       .
## other_race:priors_count           1.647782e-02
## other_race:felony                 1.394422e-02
## other_race:juv_fel_count         -3.977307e-03
## other_race:juv_misd_count         4.080779e-01
## other_race:juv_other_count       -1.644768e-01
## other_race:has_juv_felony        -5.716222e-01
## other_race:has_juv_misd          -6.862896e-01
## other_race:age_under_25          -3.777723e-02
## other_race:age_25_45             -7.315482e-02
## native_american:priors_count      2.116843e-02
## native_american:felony           -2.371184e-02
## native_american:juv_fel_count     5.667275e-02
## native_american:juv_misd_count    2.968025e-01
## native_american:juv_other_count -3.425949e-02
## native_american:has_juv_felony    4.365763e-01
## native_american:has_juv_misd      2.862483e-01
## native_american:age_under_25      2.747477e-01
## native_american:age_25_45         1.892453e-01
```

```
## priors_count:felony              1.193952e-04
## priors_count:juv_fel_count      -1.179082e-03
## priors_count:juv_misd_count     -3.065749e-04
## priors_count:juv_other_count     1.076960e-03
## priors_count:has_juv_felony     -5.526281e-03
## priors_count:has_juv_misd        1.947317e-04
## priors_count:age_under_25        2.603582e-02
## priors_count:age_25_45           9.792224e-03
## felony:juv_fel_count             2.748596e-02
## felony:juv_misd_count            1.852399e-02
## felony:juv_other_count           1.716179e-02
## felony:has_juv_felony           -4.472484e-02
## felony:has_juv_misd             -4.121938e-03
## felony:age_under_25             -9.289630e-03
## felony:age_25_45                 3.091927e-02
## juv_fel_count:juv_misd_count    -2.957597e-02
## juv_fel_count:juv_other_count    3.162266e-02
## juv_fel_count:has_juv_felony     1.345364e-02
## juv_fel_count:has_juv_misd      -2.788563e-03
## juv_fel_count:age_under_25       3.142700e-02
## juv_fel_count:age_25_45         -1.221178e-02
## juv_misd_count:juv_other_count   1.706586e-02
## juv_misd_count:has_juv_felony   -2.571617e-02
## juv_misd_count:has_juv_misd      6.921910e-03
## juv_misd_count:age_under_25     -1.273201e-02
## juv_misd_count:age_25_45        -8.842862e-03
## juv_other_count:has_juv_felony  -6.655421e-02
## juv_other_count:has_juv_misd    -3.578465e-02
## juv_other_count:age_under_25     1.919057e-02
## juv_other_count:age_25_45       -4.688677e-02
## has_juv_felony:has_juv_misd     -1.885695e-03
## has_juv_felony:age_under_25     -2.218784e-02
## has_juv_felony:age_25_45         8.704239e-02
## has_juv_misd:age_under_25       -6.167165e-03
## has_juv_misd:age_25_45           5.805613e-02
## age_under_25:age_25_45           .
```

No, Ridge does not set all the coefficients to zero exactly, although it is close. We can see that all the values have SHRUNK to near-zero. Since Ridge uses the squared values of coefficients, it shrinks values smoothly towards zero, but the nature of using Ridge means the small coefficients will never be equal to zero.

LASSO Regression: Shrink and Select

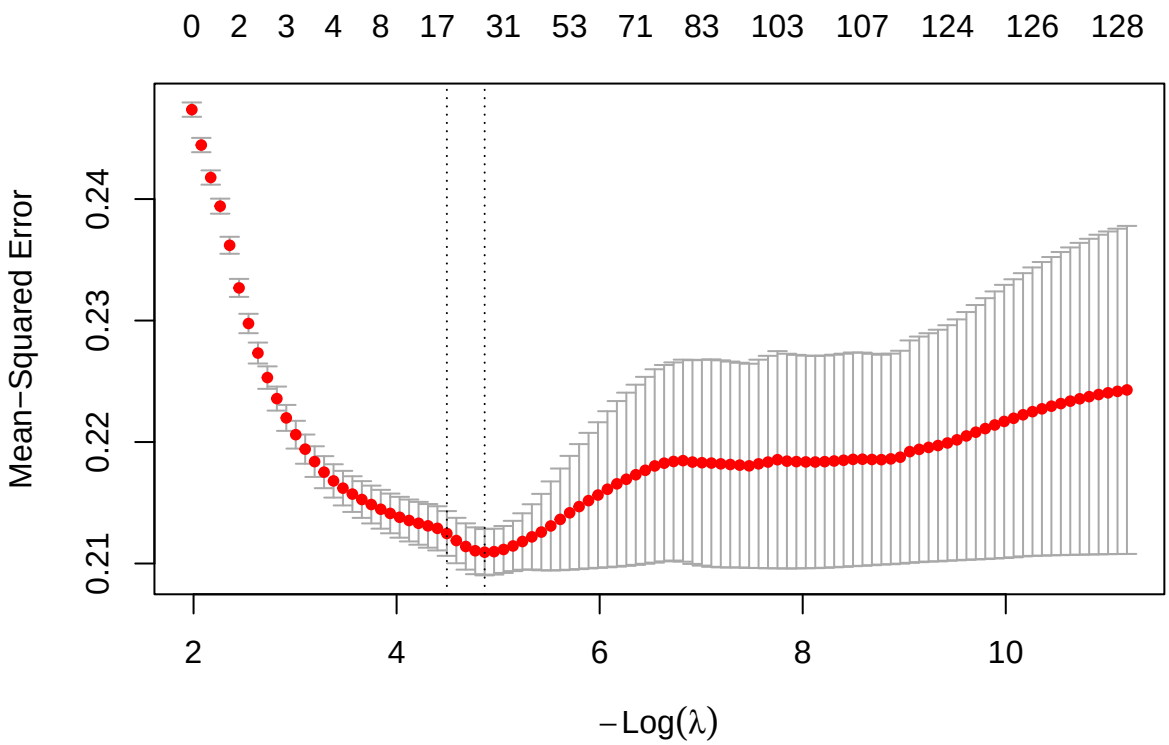
5.1 Fit LASSO With Cross-Validatoin

```r
set.seed(51)
cv_lasso <- cv.glmnet(X_train, y_train, alpha = 1, nfolds = 10)
cv_lasso$lambda.min # (a)
```

```
## [1] 0.007694477
```

(a) The optimal lambda is 0.007694477.
(b) plotted below:

```
plot(cv_lasso) #(b)
```



5.2 LASSO Variable Selection

```
lasso_coefficient <- coef(cv_lasso, s = "lambda.min")

sum(coef(cv_lasso, s = "lambda.min")[-1] != 0) # (a)
```

```
## [1] 27
```

```
coef(cv_lasso, s = "lambda.min") # (b)
```

```
## 142 x 1 sparse Matrix of class "dgCMatrix"
##                                    lambda.min
## (Intercept)                      5.968656e-01
## age                             -7.723219e-03
## female                          -5.767754e-03
## black                            .
## hispanic                         .
## asian                            .
## other_race                       .
## native_american                  .
```

```
## priors_count                     3.556519e-02
## felony                           2.138462e-02
## juv_fel_count                    .
## juv_misd_count                   .
## juv_other_count                  .
## has_juv_felony                   .
## has_juv_misd                     .
## age_under_25                     .
## age_25_45                        .
## I(age^2)                         .
## I(priors_count^2)               -4.337116e-04
## I(juv_fel_count^2)               .
## I(juv_misd_count^2)              .
## I(juv_other_count^2)            -2.220303e-03
## age:female                      -1.274802e-04
## age:black                        .
## age:hispanic                     .
## age:asian                        .
## age:other_race                   .
## age:native_american              .
## age:priors_count                 .
## age:felony                       .
## age:juv_fel_count                .
## age:juv_misd_count               .
## age:juv_other_count              .
## age:has_juv_felony               .
## age:has_juv_misd                 .
## age:age_under_25                 .
## age:age_25_45                   -8.609394e-04
## female:black                    -1.955202e-02
## female:hispanic                  .
## female:asian                     .
## female:other_race                .
## female:native_american           .
## female:priors_count              1.288987e-05
## female:felony                    .
## female:juv_fel_count             .
## female:juv_misd_count            .
## female:juv_other_count           .
## female:has_juv_felony            .
## female:has_juv_misd              .
## female:age_under_25             -9.284022e-02
## female:age_25_45                 .
## black:hispanic                   .
## black:asian                      .
## black:other_race                 .
## black:native_american            .
## black:priors_count               .
## black:felony                     .
## black:juv_fel_count              1.150518e-03
## black:juv_misd_count             .
## black:juv_other_count            3.342919e-02
## black:has_juv_felony             .
## black:has_juv_misd               .
```

```
## black:age_under_25               7.940064e-02
## black:age_25_45                  .
## hispanic:asian                   .
## hispanic:other_race              .
## hispanic:native_american         .
## hispanic:priors_count            .
## hispanic:felony                  .
## hispanic:juv_fel_count           9.218711e-02
## hispanic:juv_misd_count          .
## hispanic:juv_other_count         .
## hispanic:has_juv_felony          .
## hispanic:has_juv_misd            .
## hispanic:age_under_25            .
## hispanic:age_25_45              -8.940017e-03
## asian:other_race                 .
## asian:native_american            .
## asian:priors_count               .
## asian:felony                     .
## asian:juv_fel_count              .
## asian:juv_misd_count             .
## asian:juv_other_count            .
## asian:has_juv_felony             .
## asian:has_juv_misd               .
## asian:age_under_25               .
## asian:age_25_45                 -2.166967e-02
## other_race:native_american       .
## other_race:priors_count          1.547459e-04
## other_race:felony                .
## other_race:juv_fel_count         .
## other_race:juv_misd_count        .
## other_race:juv_other_count       .
## other_race:has_juv_felony       -3.579049e-02
## other_race:has_juv_misd          .
## other_race:age_under_25          .
## other_race:age_25_45            -1.748625e-02
## native_american:priors_count     .
## native_american:felony           .
## native_american:juv_fel_count    .
## native_american:juv_misd_count   .
## native_american:juv_other_count  .
## native_american:has_juv_felony   .
## native_american:has_juv_misd     .
## native_american:age_under_25     .
## native_american:age_25_45        .
## priors_count:felony              .
## priors_count:juv_fel_count       .
## priors_count:juv_misd_count      .
## priors_count:juv_other_count     .
## priors_count:has_juv_felony      .
## priors_count:has_juv_misd        .
## priors_count:age_under_25        1.682224e-02
## priors_count:age_25_45           2.040836e-03
## felony:juv_fel_count             .
## felony:juv_misd_count            .
```

```
## felony:juv_other_count           .
## felony:has_juv_felony            .
## felony:has_juv_misd              4.124298e-03
## felony:age_under_25              .
## felony:age_25_45                 .
## juv_fel_count:juv_misd_count     .
## juv_fel_count:juv_other_count    .
## juv_fel_count:has_juv_felony     .
## juv_fel_count:has_juv_misd       .
## juv_fel_count:age_under_25       7.147058e-03
## juv_fel_count:age_25_45          .
## juv_misd_count:juv_other_count   .
## juv_misd_count:has_juv_felony   -2.591047e-03
## juv_misd_count:has_juv_misd      .
## juv_misd_count:age_under_25      .
## juv_misd_count:age_25_45         .
## juv_other_count:has_juv_felony   .
## juv_other_count:has_juv_misd     .
## juv_other_count:age_under_25     1.918434e-02
## juv_other_count:age_25_45        .
## has_juv_felony:has_juv_misd      .
## has_juv_felony:age_under_25      8.816157e-03
## has_juv_felony:age_25_45         .
## has_juv_misd:age_under_25        .
## has_juv_misd:age_25_45           .
## age_under_25:age_25_45           .
```

(a) Among the 141 predictors, 27 variables were retained and have non-zero coefficients, while 114 variables shrunk to zero/were eliminated.
(b) Some of the surviving interaction terms are age:female, female:black, or like hispanic:age_25_45. I think this makes intuitive sense because they show that the effect of things like prior offenses or gender can change based on gender/race. This is realistic, since recidivism is likely influenced by a combination of factors, and not just each one acting on its own.
(c) LASSO selects the variables to drop based on predictive values and not arbitrary measures, so it only includes the relevant predictors. This makes it a more objective model than if, for example, a researcher were to randomly decide what interactions to include or not.

5.3 LASSO RMSE

```r
lasso_predicted <- predict(cv_lasso, newx = X_test, s = "lambda.min")
rmse_lasso <- sqrt(mean((lasso_predicted - y_test)^2))
rmse_lasso
```

```
## [1] 0.4592597
```

The LASSO RMSE is 0.4592597, which is lower than the simple OLS of 0.462227, kitchen-sink OLS of 0.4890411, and Ridge of 0.4700385. This tells us that LASSO performs the best prediction because it shrinks the coefficients but still eliminates the irrelevant variables, so it's a good tool to use to avoid overfitting.

6 Model Comparison and Interpretation

6.1 Elastic Net

```
set.seed(51) # run given code first
cv_enet <- cv.glmnet(X_train, y_train, alpha = 0.5, nfolds = 10)

elastic_predicted <- predict(cv_enet, newx = X_test, s = "lambda.min") # RMSE

rmse_elastic <- sqrt(mean((elastic_predicted- y_test)^2))
rmse_elastic
```

```
## [1] 0.4583446
```

```
sum(coef(cv_enet, s = "lambda.min")[-1] != 0) # variables count
```

```
## [1] 43
```

The elastic net out-of-sample RMSE is 0.4583446 and it retains 43 variables.

6.2 The Big Comparison Table

```
table <- tibble(
  Model = c("OLS (main effects, 16 vars)",
            "OLS (kitchen sink, ~141 vars)",
            "Ridge",
            "LASSO",
            "Elastic Net"),

  Predictors = c(16,
                 141,
                 141,
                 27,
                 43),

  RMSE = c(0.462227,
           0.4890411,
           0.4700385,
           0.4592597,
           0.4583446)
)

kable(table)
```

Model	Predictors	RMSE
OLS (main effects, 16 vars)	16	0.4622270
OLS (kitchen sink, ~141 vars)	141	0.4890411

Model	Predictors	RMSE
Ridge	141	0.4700385
LASSO	27	0.4592597
Elastic Net	43	0.4583446

(a) The elastic net model has the lowest out-of-sample RMSE at 0.4583446.
(b) The kitchen-sink model has the best sample fit because it has the most predictors (141) so it is able to minimize training error. That being said, this model is not necessarily better for out-of-sample data because it overfits.
(c) What the table tells us is that adding more predictors/interactions improves in-sample fit, but it does not translate to better out-of-sample performance. We see that in (b) with the kitchen-sink model, with high variance. On the other hand, LASSO and elastic net introduce slight bias but does reduce variance in data, which is where the bias-variance tradeoff comes in.

6.3 Baker Article Connection

(a) LASSO was particularly useful in the Halliburton case because it allowed for objective assessment of firms that best predicts the stock's returns. Rather than the subjective hand-selected approach of deciding which firms to include, LASSO uses a penalized approach, so it selected firms based on predictive approaches that drop firms that do not improve the predictions, which settles the disagreement by grounding the model in objective data.
(b) Based on my own experience, and Baker's explanation, the OLS model allows for a "bespoke" approach in that we base what to include on our judgment, deciding what variables to manipulate or not. On the other hand, LASSO eliminates this discretion of choice by automatically omitting irrelevant predictors.

6.4 Prediction vs Manipulation

(a) No, LASSO is not the tool for causal questions because it is designed to improve prediction and may end up shrinking coefficients.
(b) The difference is that regression for prediction seeks to make the most accurate predictions of outcomes, like recidivism, while causal inference asks something like whether prior offenses actually CAUSE recidivism, and the research design is important in ensuring we can isolate the cause and effect.

6.5 Algorithmic Fairness

```r
white <- with(test, black == 0 & hispanic == 0 & asian == 0 & # white defendants
                    other_race == 0 & native_american == 0)

sum(test$black == 1)   # count black defendants
```

```
## [1] 729
```

```r
sum(white)             # count white defendants
```

```
## [1] 495
```

```r
mean(test_data[test$black == 1])   # (a) predicted
```

```
## [1] 0.511985
```

```r
mean(test_data[white])
```

```
## [1] 0.3789606
```

```r
mean(test$recidivism[test$black == 1])  # (b) actual
```

```
## [1] 0.521262
```

```r
mean(test$recidivism[white])
```

```
## [1] 0.4020202
```

(a) The mean predicted recidivism rates are 0.511985 for black defendants and 0.3789606 for white defendants, or 51.1985% and 37.89606% respectively. There is a difference, which is that the model predicts a higher recidivism likelihood for black defendants.

(b) The actual recidivism rate is 0.521262 or 52.1262% for black defendants, and 0.4020202, or 40.20202% for white defendants. These values are both higher than what the model predicted, but, the predictive model is ultimately still able to predict the fact that black recidivism rates are higher than white.

(c) I think a potential tradeoff to consider is that, while we are able to reduce human discretion in criminal justice decisions by observing objective data, we must do so looking at historical data, which could result in some biases. Additionally, the predictive method relies on methods that are correlated with outcomes, so they're not truly causal. It would be difficult to justify in the decision-making processes if we cannot establish and isolate for a causal relationship (ex: race causes recidivism).